



SnapRoute

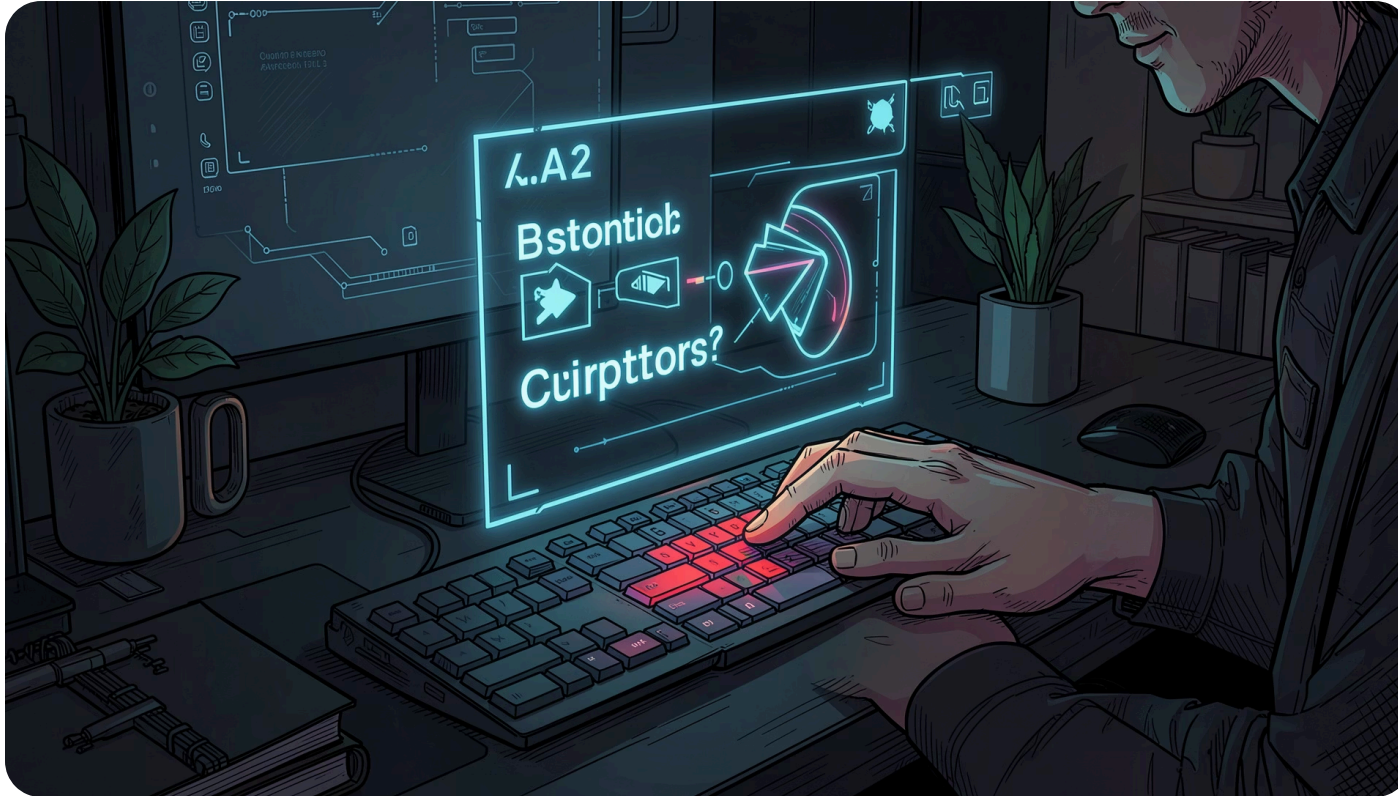
A premium, ultra-lightweight, 100% portable Windows screen capture, OCR, and visual search utility — triggered by a single global hotkey.

PYTHON 3 + PYQT6

STANDALONE .EXE

WINDOWS DESKTOP

What Is SnapRoute?



SnapRoute freezes the display with a cinematic fade-in, lets you crop any region with a glowing neon border, and presents an instant HUD overlay that smart-routes your selection.

Copy Text

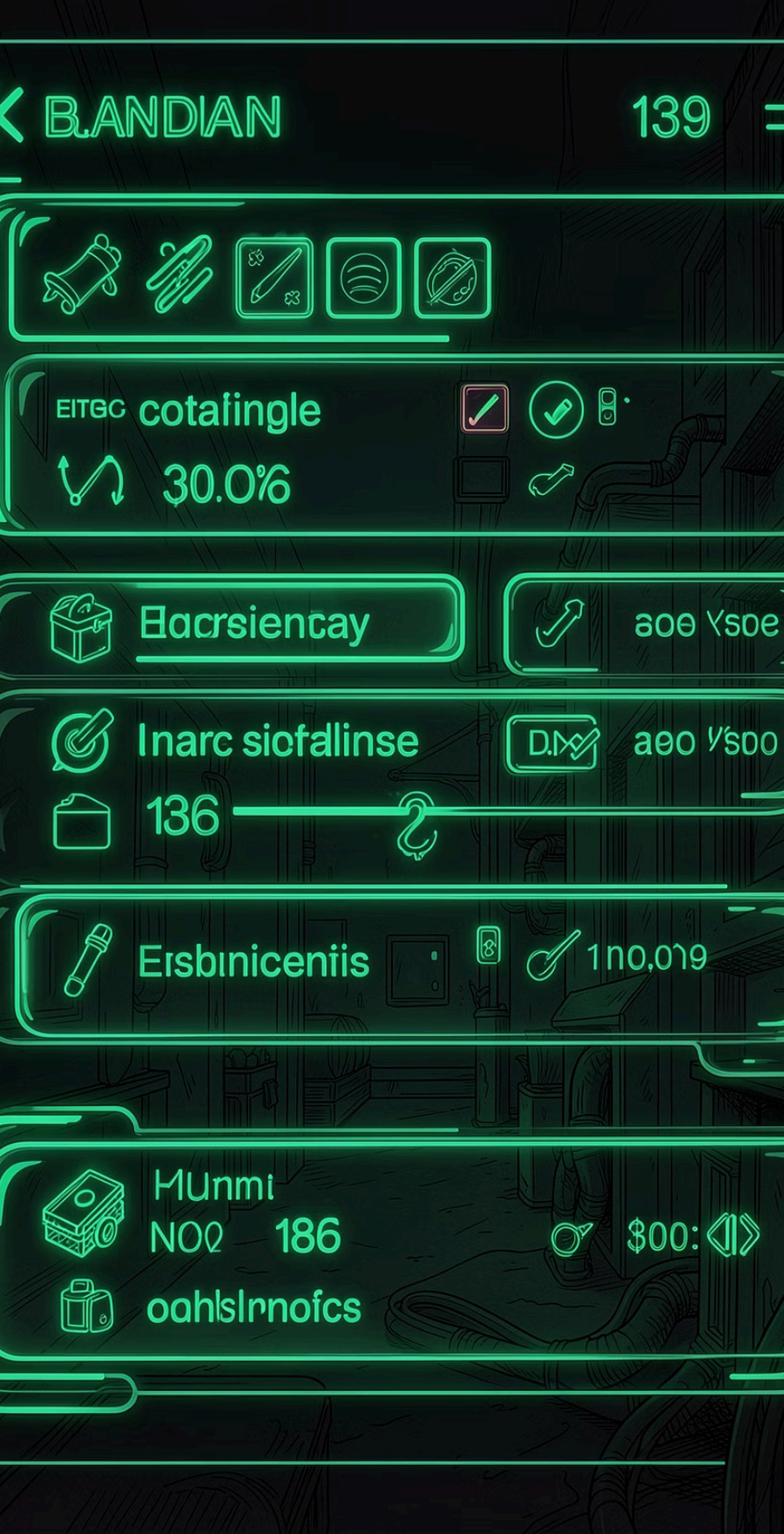
Extracted via OCR to clipboard

Search Text

Launches Google search instantly

Search Image

Uploads to ephemeral host → Google Lens



Key Features



Cinematic Dim Fade-in

Display freezes and dim-fades into a dark navy translucent overlay via multi-threaded frame grabber (80 FPS).



High-Visibility Cursor

Custom neon-emerald crosshair with dark drop-shadows for visibility on any background.



Glassmorphic HUD

Translucent dock overlay with context-aware action buttons and HSL glassmorphism styles.



Context-Aware Smart Routing

Auto-detects readable characters. If ≥ 2 found, "Search Text" is enabled dynamically.

Zero-Configuration Portability

Single Executable Delivery

SnapRoute requires **no Python runtime** and no external installations. Everything — including the Tesseract-OCR binary engine — is fully bundled inside one portable .exe.

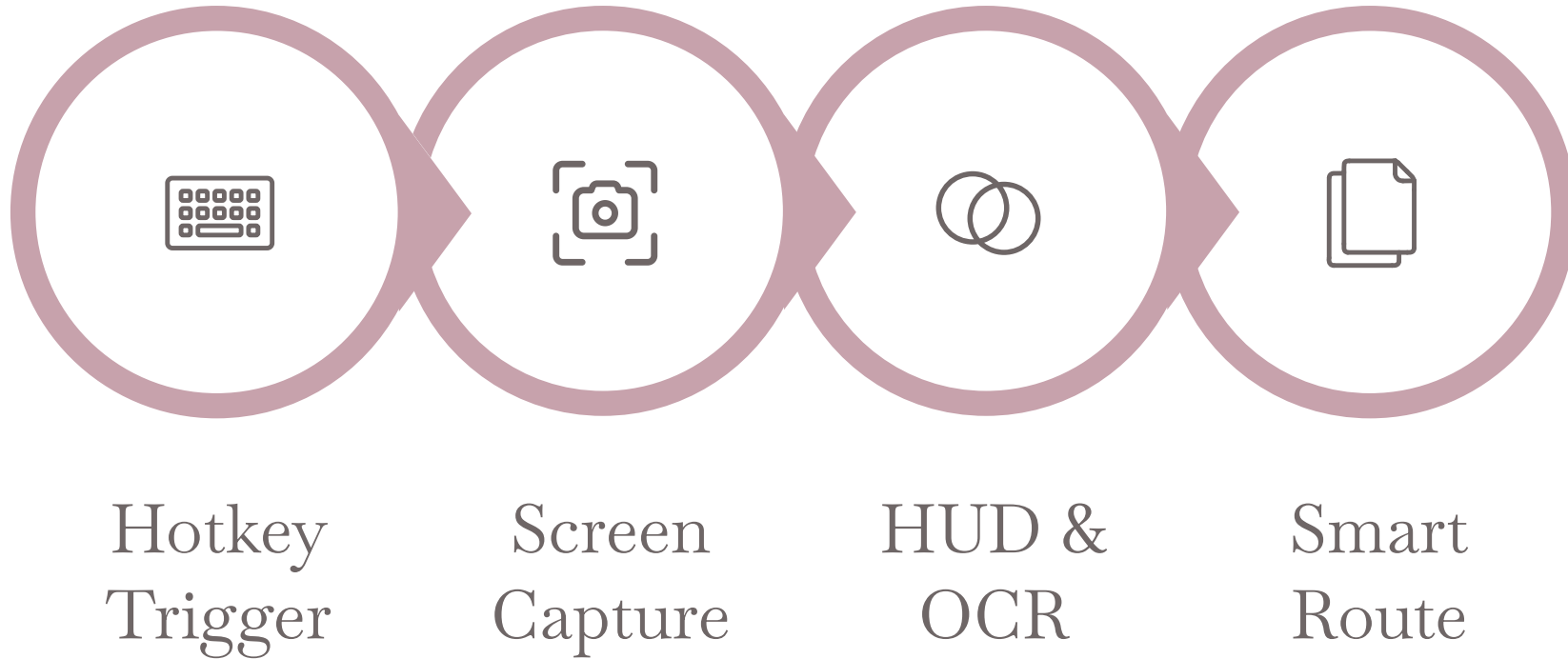
- ❶ Built with PyInstaller, bundling portable Tesseract-OCR asset folder mappings for true zero-dependency distribution.

Head font:
Baskerville.

The Quick
Body font: Robbins Use
Brown Fox it for paragraphs and
Jumps Over smaller text.

ABCDEFGHIJKLMNOPQRSTUVWXYZ
This is the accent color, use
abcdefghijklmnopqrstuvwxyz
0123456789.
decorations

Execution Flow



All blocking operations — OCR, network uploads, and keyboard hook monitoring — are offloaded to background threads, keeping the main GUI thread at **0ms delay**.

Core Components

01

System Tray Daemon (main.py)

Starts the event loop, creates the taskbar tray menu, triggers the splash animation, and spawns the global hotkey listener.

02

Keyboard Polling Manager

Monitors a thread-safe `queue.Queue` from a background thread to intercept `Ctrl + Shift + S` without blocking the Qt thread.

03

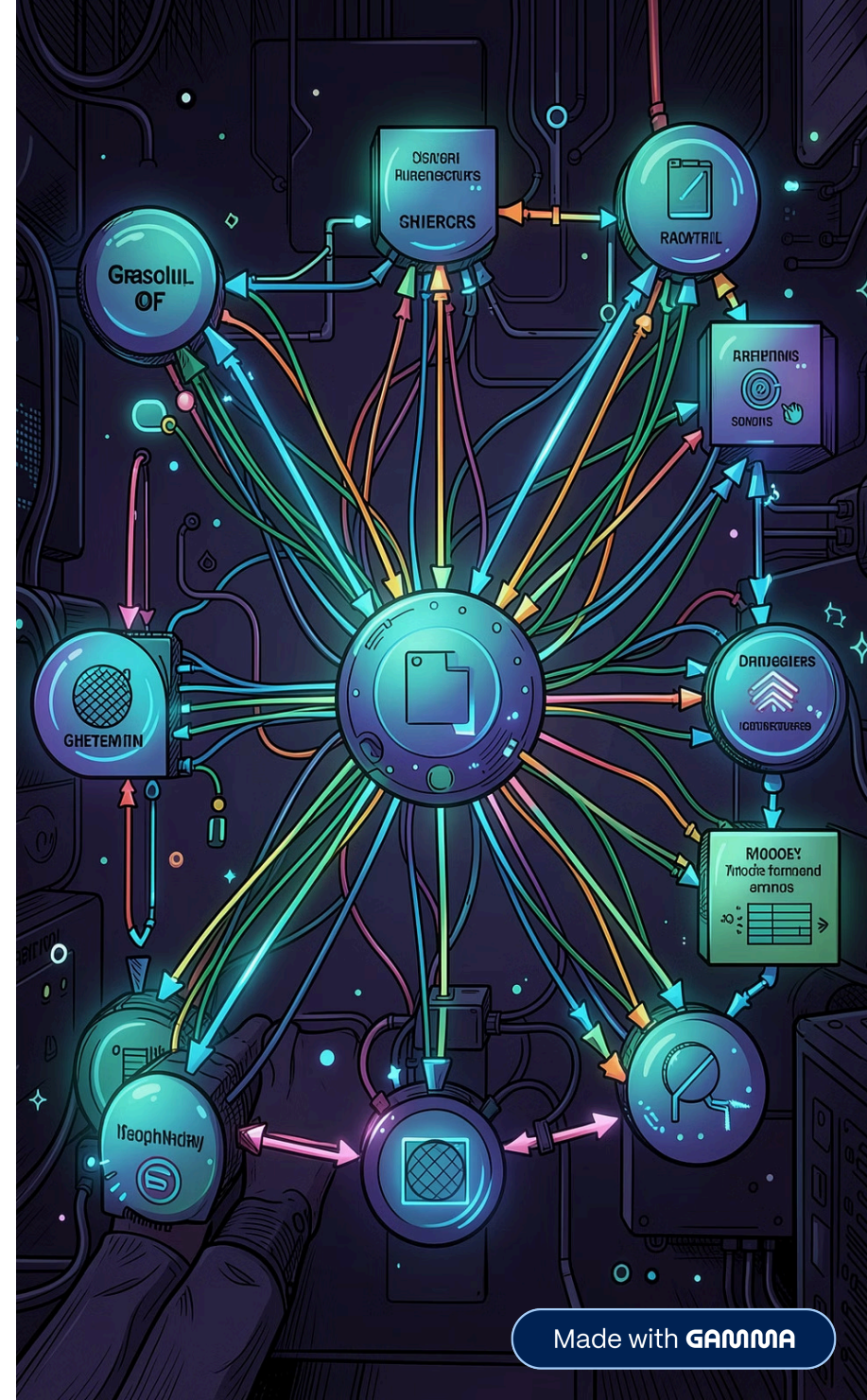
Screen Overlay UI (box_search.py)

Creates the frameless fullscreen window, captures the display screenshot, and controls HUD positioning.

04

OCRWorker & UploadWorker Threads

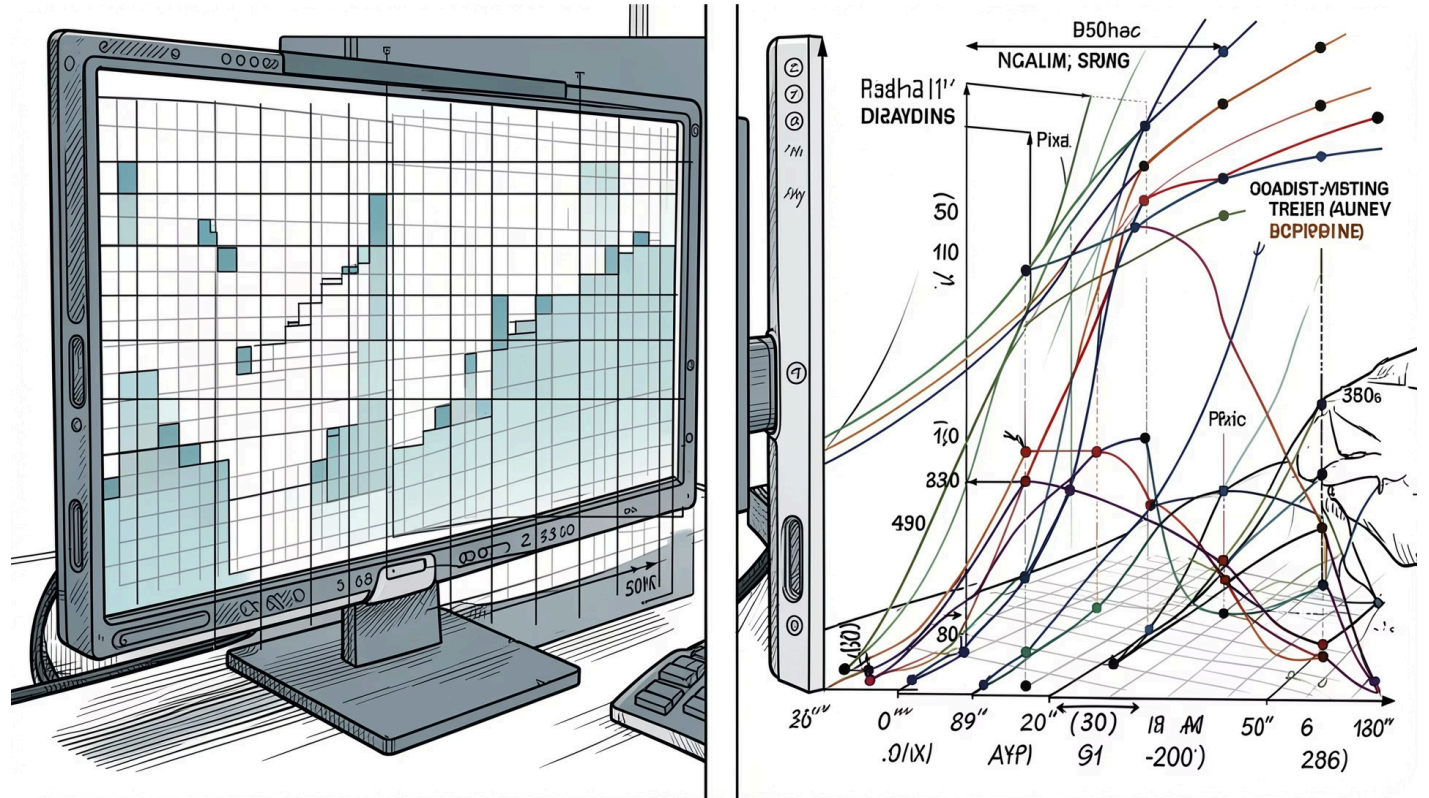
Background threads run `pytesseract` and sequential image uploads to prevent main window freeze.



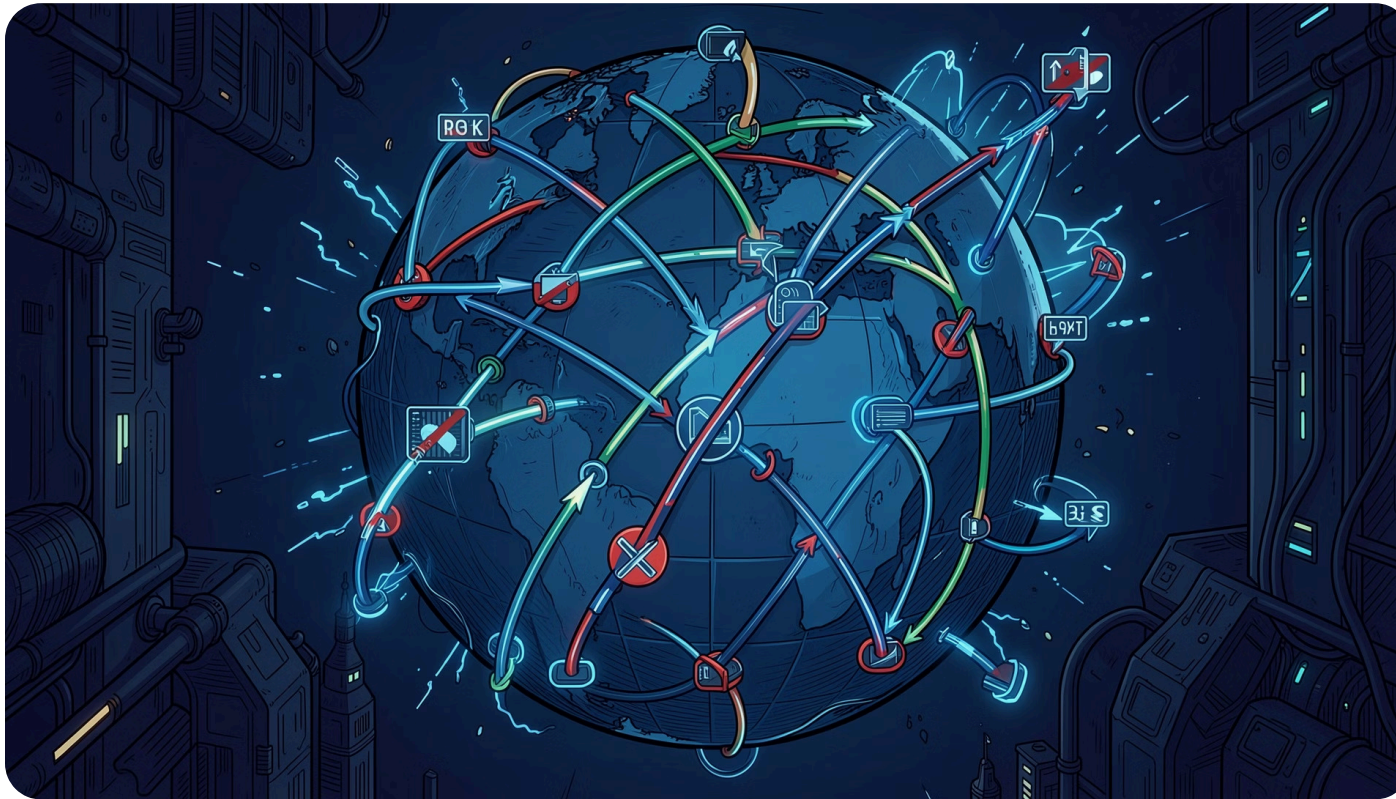
Windows High-DPI Scaling Mismatch

⚠ **The Problem:** On 4K/2K displays with Windows scaling (125%–150%), mouse events return **logical pixels** while screenshots are captured in **physical pixels**. Manual DPI multipliers caused double-scaling, yielding blank or offset crops.

✅ **The Solution:** Removed manual DPI multipliers entirely. PyQt6's `QPixmap.copy` handles logical-to-physical coordinate mapping natively — crops became **100% pixel-perfect**.



ISP/DNS-Level Image Host Blocking



The Problem: Hosts like Catbox and Pixeldrain are blocked at the DNS/ISP level by providers like Jio and Airtel in India, causing upload timeouts.

The Solution: A multi-host sequential upload worker attempts uploads in order:

1

Uguu.se

Primary — globally unblocked, deletes files after 24h

2

Catbox.moe

Fallback #1

3

Pixeldrain.com

Fallback #2

HUD Jitter & Window Hierarchy Conflicts

HUD Button Jitter

Problem: Async OCR completion caused the HUD to resize and re-center, shifting buttons under the cursor.

Fix: HUD uses a fixed 440px width. "Search Text" starts in a low-opacity **disabled state** and is simply enabled when OCR completes — zero layout shift.

Stays-on-Top Conflict

Problem: Triggering the hotkey right after launch caused the Welcome splash to overlap the crop overlay.

Fix: Startup logic detects if the splash is active during a hotkey trigger and instantly closes it, avoiding coordinate and window-level conflicts.



UI/UX Design & Aesthetics

Glassmorphism

HUD uses `rgba(12,12,20,245)`
backgrounds with neon-emerald
`rgba(0,255,136,200)` borders.

Multi-Layer Glow Borders

Three-layer crop rectangle: 8px outer
glow, 4px accent glow, 2px ultra-bright
core.

Opacity Animations

`QPropertyAnimation` on `windowOpacity`: **180ms** fade-in, **150ms** fade-out — fluid and
jitter-free.